

Auteur: Eryk Borczuch  
Studierichting: Informatica  
Oefeningenreeks 4A nr. 15.5

$$\begin{aligned} & \int (x^2 - x - 4)e^x dx \\ & \left\{ \begin{array}{l} u = x^2 - x - 4 \\ dv = e^x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 2x - 1 dx \\ v = e^x \end{array} \right. \\ & = (x^2 - x - 4)e^x - \int (2xe^x - e^x) dx \\ & \left\{ \begin{array}{l} u = 2x \\ dv = e^x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 2 dx \\ v = e^x \end{array} \right. \\ & = (x^2 - x - 4)e^x - 2xe^x - 2e^x + 2e^x + e^x \\ & = e^x(x^2 - 3x - 1) + c \end{aligned}$$

## References